

केन्द्रीय विद्यालय संगठन, कोलकाता संभाग
KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION
प्री-बोर्ड परीक्षा / PRE – BOARD EXAMINATION – 2025-26

कक्षा /CLASS - X
विषय /SUBJECT - SCIENCE

अधिकतम अंक /MAXIMUM MARKS - 80
समय /TIME – 3 घंटे / 3 HOURS

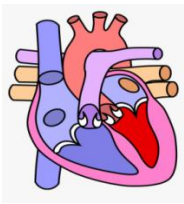
GENERAL INSTRUCTIONS

- i) This question paper consists of 39 questions in three sections. Section A is Biology, Section B is chemistry and Section C is Physics.
- ii) All questions are compulsory . However , an internal choice is provided in some questions . A student is expected to attend only one of these questions

SECTION – A

Q.NO	QUESTIONS	MARKS
1	In the human respiratory system , the path taken by air when we breathe in is : a) Nostrils – Larynx – Trachea – Pharynx – Alveoli b) Nostrils – Trachea – Larynx – Pharynx – Alveoli c) Nostrils – Pharynx – Larynx – Trachea – Alveoli d) Nostrils – Larynx – Pharynx – Trachea – Alveoli	1
2	Parasitic mode of nutrition is observed in : a) Bryophyllum b) Hibiscus c) Cuscuta d) Helianthus	1
3	Which of the following statements are true about the brain ? i) The main thinking part of the brain is hind brain ii) Centres of hearing , smell , memory , sight etc are located in forebrain. iii) Involuntary actions like salivation , vomiting , blood pressure are controlled by the medulla in the hind-brain iv) Cerebellum does not control posture and balance of the body. a) (i) and (ii) b) (i) , (ii) and (iii) c) (iii) and (iv) d) (ii) and (iii)	1

4	In a nerve cell , the site where the electrical impulse is converted into a chemical signal is known as : a) Axon b) Neuromuscular junction c) Dendrites d) Cellbody	1
5	When a pure tall pea plant is crossed with a pure dwarf pea plant , the percentage of tall pea plant in F ₁ and F ₂ generation pea plants will be respectively : a) 100% ; 25% b) 100% ; 50% c) 100% ; 75% d) 100% ; 100%	1
6	Which are the correct statements related to Ozone ? i)Ozone layer helps in increasing the UV radiations reaching Earth . ii)Ozone is a deadly poison. iii)Ozone layer shields the earth from UV radiations. iv)Ozone layer prevents UV rays which cause skin cancers. v)Ozone is formed with the help of Chlorofluro carbons a) (i) , (ii) and (iii) b) (ii) , (iii) and (iv) c) (iii) , (iv) and (v) d) (i) , (iv) and (v)	1
7	An ecosystem comprising of producers , herbivores and carnivores is an incomplete ecosystem if it does not include : a) Secondary consumers b) Decomposers c) Top carnivores d) Secondary and tertiary consumers	1
	Question numbers 8 and 9 consists of two statements – Assertion (A) and Reason (R) . Answer these questions by selecting the appropriate option given below a) Both A and R are true, and R is the correct explanation of A b) Both A and R are true , and R is not the correct explanation of A c) A is true but R is false d) A is false but R is true	
8	Assertion : In reptiles , the temperature at which the fertilized eggs are kept decides the sex of the offsprings. Reason : Sex is not genetically determined in some animals.	1

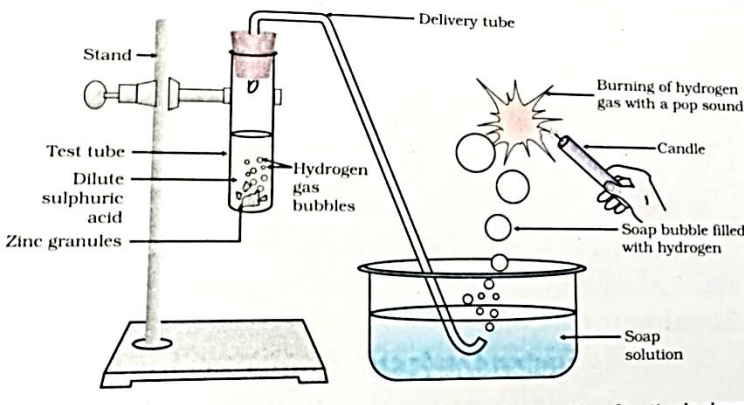
9	Assertion : All energy from the producers is utilised by the consumers . Reason : Only about 10% of energy available to a trophic level is transferred to the next higher trophic level.	1
10	Explain in brief two ways by which leaves of a plant help in excretion .	2
11	Students to attempt either option A or B A) Name the chamber that receives oxygen rich blood and name the blood vessel which brings it.  OR B)What will happen if : i)Xylem tissue in a plant is removed ? ii)We are injured and start bleeding ?	2
12	If 10000 J energy is available at the producer level , how much energy will be available to the secondary consumers ?	2
13	Define reflex action. With the help of a flow chart show the path of a reflex action such as sneezing.	3
14	A blue coloured flower plant denoted by BB is crossed - bred with that of white coloured flower plant denoted by bb. i)State the colour of flower you would expect in their F_1 generation plants. ii)What must be the percentage of white flower plants in F_2 generation if flowers of F_1 plants are self - pollinated. iii)State the expected ratio of the genotypes BB and Bb in the F_2 progeny.	3
15	Kidneys are vital organs for survival. Several factors like infections, injury or restricted blood flow to kidneys reduce the activity of kidneys. This leads to accumulation of poisonous wastes in the body, which can even lead to death. In case of kidney failure, an artificial kidney can be used. An artificial kidney is a device to remove waste products from the blood through dialysis i) (a)Name the artery that brings oxygenated blood to the kidney. (b)Name the cluster of thin-walled blood capillaries present in the Bowman's capsule ii) In human excretory system name the organ which stores urine. Is this organ under hormonal control or nervous control ?	4

	iii) List two major steps involved in the formation of urine and state in brief their functions OR iii) In which part of the nephron does selective reabsorption takes place? List the factors on which the amount of water reabsorbed depends on.	
16	i) Name the parts of a bisexual flower that are not directly involved in reproduction. ii) Differentiate between self pollination and cross pollination. List any two significance of pollination. iii) What is the fate of ovules and ovary after fertilisation in a flower? OR i) Where are testes located in the human males and why? State two function of the testes ii) In the human female, one of the ovaries releases an egg every month. State the changes that take place if (a) The egg is fertilised, and (b) The egg is not fertilised. iii) What is done during the surgical method in males and females to prevent pregnancy?	5

SECTION – B

17	Which of the following is displacement as well as redox reaction ? a) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ b) $2\text{FeSO}_4 \xrightarrow{\text{heat}} \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$ c) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ d) $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$	1
18	Metal oxides reacts generally with acids but few oxides of metal also react with bases. Such metallic oxides are (i) MgO (ii) ZnO (iii) Al_2O_3 (iv) CaO a) (i) and (ii) b) (ii) and (iii) c) (iii) and (iv) d) (i) and (iv)	1
19	Can we store iron sulphate(aq) in copper container? (a) Yes, because iron is more reactive than copper. (b) No, because iron is less reactive than copper. (c) Yes, because copper is more reactive than iron. (d) No, because copper is less reactive than zinc.	1

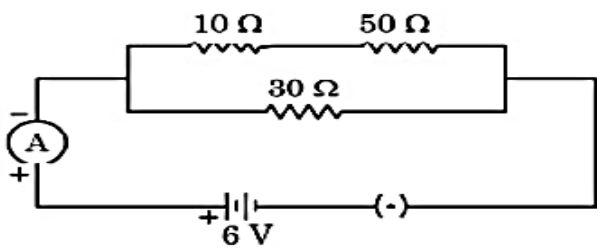
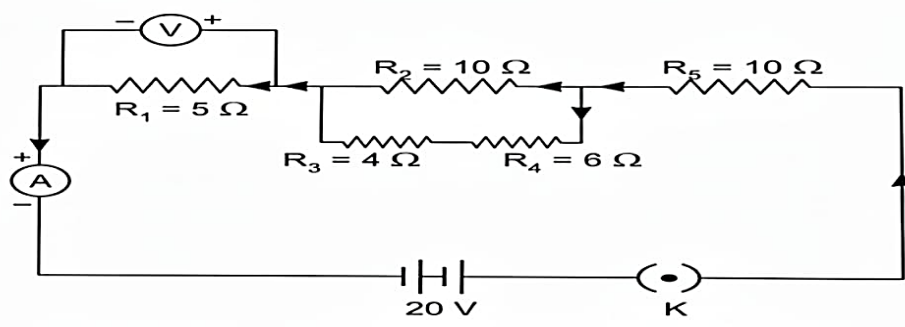
20	Methyl orange is added to dilute hydrochloric acid and to aqueous sodium hydroxide. What is the colour of the methyl orange in each solution		1
	sample	Colour in dilute hydrochloric acid	
	A	Orange	
	B	Red	
	C	Red	
	D	Yellow	
21	1 M HCl has lower pH than 1 M CH ₃ COOH because (a) HCl is a weak acid (b) HCl is strong acid, completely ionised (c) Acetic acid is weak acid, partially ionised (d) Both (b) and (c)		1
22	Reema took 5 ml of lead nitrate solution in a beaker and added approximately 4 ml of KI solution to it. What would she observe? (a) The solution turned red (b) Yellow precipitate was formed (c) White precipitate was formed (d) The reaction mixture becomes hot		1
23	A student took sodium sulphate solution in a test tube and added barium chloride solution to it. He observed that an insoluble substance has formed. The colour and molecular formula of the insoluble substance is: (a) Grey, Ba ₂ SO ₄ (b) Yellow, Ba(SO ₄) ₂ (c) White, BaSO ₄ (d) Pink, BaSO ₄		1
	Question number 24 consists of two statements – Assertion (A) and Reason (R) . Answer these questions by selecting the appropriate option given below a) Both A and R are true, and R is the correct explanation of A b) Both A and R are true, and R is not the correct explanation of A c) A is true but R is false d) A is false but R is true		
24	Assertion : Ethene is more reactive than Ethane. Reason : Ethene has a double bond, while Ethane has a single bond between carbon atoms.		1
25	“Displacement reactions also play a key role in extracting metals in the middle of the reactivity series”. Justify this statement with two examples.		2

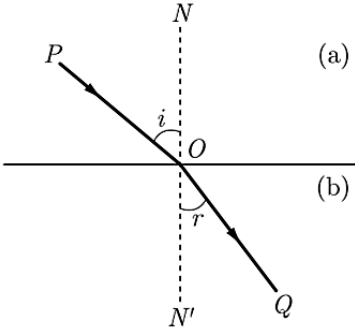
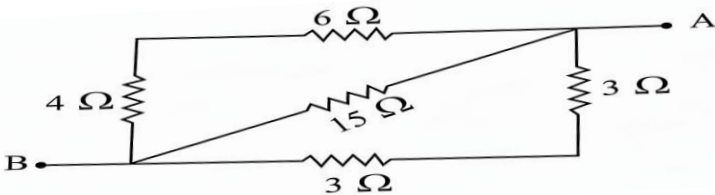
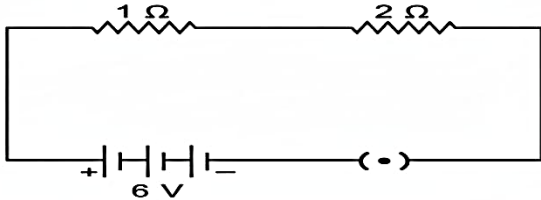
26	<table><tr><th>Column I</th><th>Column II</th></tr><tr><td>(a)Electrolytic reduction</td><td>Al</td></tr><tr><td></td><td>Zn</td></tr><tr><td>(b)Reduction with carbon</td><td>Na</td></tr><tr><td></td><td>Fe</td></tr><tr><td></td><td>Sn</td></tr></table> (i)Which one of the methods given in column I are used for extraction of each of the metals given in column II (ii) Draw the Lewis dot structure for $MgCl_2$ OR (i)An ore on heating in air produces sulphur dioxide. Which process would you suggest for its concentration? (ii) Describe briefly any two steps involved in the conversion of this concentrated ore into related metal	Column I	Column II	(a)Electrolytic reduction	Al		Zn	(b)Reduction with carbon	Na		Fe		Sn	3
Column I	Column II													
(a)Electrolytic reduction	Al													
	Zn													
(b)Reduction with carbon	Na													
	Fe													
	Sn													
27	A student named Rakhi was performing an experiment in chemistry lab. The following diagram shows the reaction of zinc granules with dilute sulphuric acid and testing hydrogen gas by burning.  What would happen if the following changes are made? (a) In place of zinc granules, same amount of zinc dust is taken. (b) In place of zinc granules, copper pieces are taken (c) Instead of dilute sulphuric acid, nitric acid is taken.	3												
28	During a chemistry lab session , Minu accidentally spilled a small quantity of hydrochloric acid on the table . The lab assistant responded quickly by sprinkling baking soda on the spill , which caused brisk effervescence . observing this , the chemistry teacher explained the chemical reaction that occurred and emphasised on the importance of following safety protocols when working with acids ain the lab. a)Name the type of reaction and write the name of the products formed in the reaction. b)Why did the lab assistant sprinkled baking soda over the spill ? OR b)List two safety measures that the teacher might have explained to the students while handling acids . c)What caused the evolution of brisk effervescence and how can you confirm it ? Write a balanced chemical equation.	4												

29	A carbon compound 'A' is widely used as a preservative in pickles and has a molecular formula $C_2H_4O_2$. This compound reacts with ethanol to form a sweet smelling compound 'B'. (i) Identify the compound 'A' and write its structure. (ii) Write the chemical equation for the reaction of 'A' with ethanol to form compound 'B'. State the role of presence of an acid in the reaction. (iii) How can we get compound 'A' back from 'B'? (iv) How can 'A' be obtained from ethanol? (v) Name the gas produced when compound 'A' reacts with washing soda. OR (i) Name a commercially important carbon compound having functional group $-OH$ and write its molecular formula. (ii) Write chemical equation to show its reaction with (a) Sodium metal (b) Excess conc. Sulphuric acid (c) Ethanoic acid in the presence of an acid catalyst (d) Acidified potassium dichromate Also write the name of the product formed in each case.	5
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SECTION – C

30	The nature of image formed by a convex mirror when the object distance from the mirror is less than the distance between pole and focal point 'F' of the mirror would be (a) Real, inverted and diminished in size (b) Real, inverted and enlarged in size (c) Virtual , upright and diminished in size (d) Virtual, upright and enlarged in size	1
31	Twinkling of stars is due to atmospheric (a) Dispersion of light by water droplets (b) Refraction of light by different layers of varying refractive indices (c) Scattering of light by dust particles (d) Internal reflection of light by clouds	1
	Question number 32 consists of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below (a)Both A and R are true, and R is the correct explanation of A (b) Both A and R are true , and R is not the correct explanation of A (c) A is true but R is false (d) A is false but R is true	
32	Assertion: Higher the refractive index of the medium, lesser will be the velocity of light in that medium. Reason: Refractive index is inversely proportional to the velocity of light in the medium.	1

33	(i) An object of 5 cm height is placed at a distance of 20 cm from the optical centre of a concave lens of focal length 18 cm. Calculate (a) Image distance and (b) Magnification in this case (II) Compare the values of magnification obtained by a concave lens and a convex lens when both the lenses form virtual images	2
34	In the given circuit determine the value of :  i) Total resistance of the circuit ii) Current flowing through the ammeter OR Study the following circuit and find :  i) Effective resistance of the circuit ii) Current drawn from the battery	2
35	A person is suffering from an eye- defect in which the far point of the eye is much nearer than infinity. Name the defect of vision the person is suffering from. list two main causes of this defect. write the type of the corrective lens and draw a ray diagram to show the function of the corrective lens	3
36	An electric motor rated 1100 Watt is connected to 220 V mains. Find (A) The current drawn from the mains (B) Electric energy consumed if the motor is used for 5 hours daily for 6 days (C) Total cost of energy consumed if the rate of 1 unit is rupees 5	3

37	State the rule to determine the direction of a (i) magnetic field produced around a straight conductor-carrying current, (ii) force experienced by a current-carrying straight conductor placed in a magnetic field which is perpendicular to it, and (iii) current induced in a coil due to its rotation in a magnetic field.	3
38	When light ray goes from one transparent medium to another transparent medium, it suffers a change in direction, into second medium. The extent of the change in direction suffered by the phenomenon of change in the path of light rays when going from one medium to another medium is known as refraction. Ray is a given pair of media can be expressed in terms of refractive index. The refractive index is related to an important physical quantity in the relative speed of light in different media.  (i) When light goes from one medium to another, which of the three parameters: frequency, wavelength, velocity change? (ii) A ray of light enters into the glass from air. Does it bend towards normal? OR Define Snell's law. (iii) Light enters from air to glass having refractive index 1.50. What is the speed of light in the glass? The speed of light in vacuum is $3 \times 10^8 \text{ ms}^{-1}$.	4
39	a)List two disadvantages of using a series circuit in homes. b)Calculate the effective resistance between A and B in the circuit given below :  OR a)In the given circuit calculate the power consumed in watts in the resistor of 2Ω :  b)Compare the power used in the 2Ω resistor in each of the following circuits : i) a 6V battery in series with 1Ω and 2Ω resistors , and ii) a 4V battery in parallel with 12Ω and 2Ω resistors .	5

